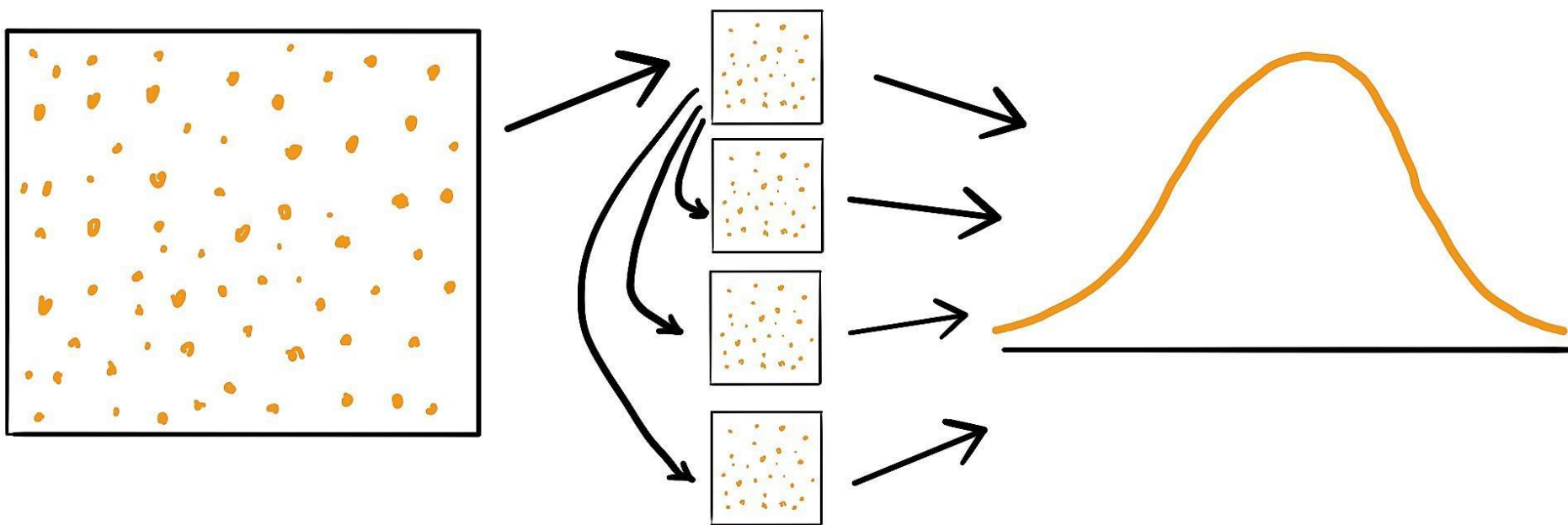
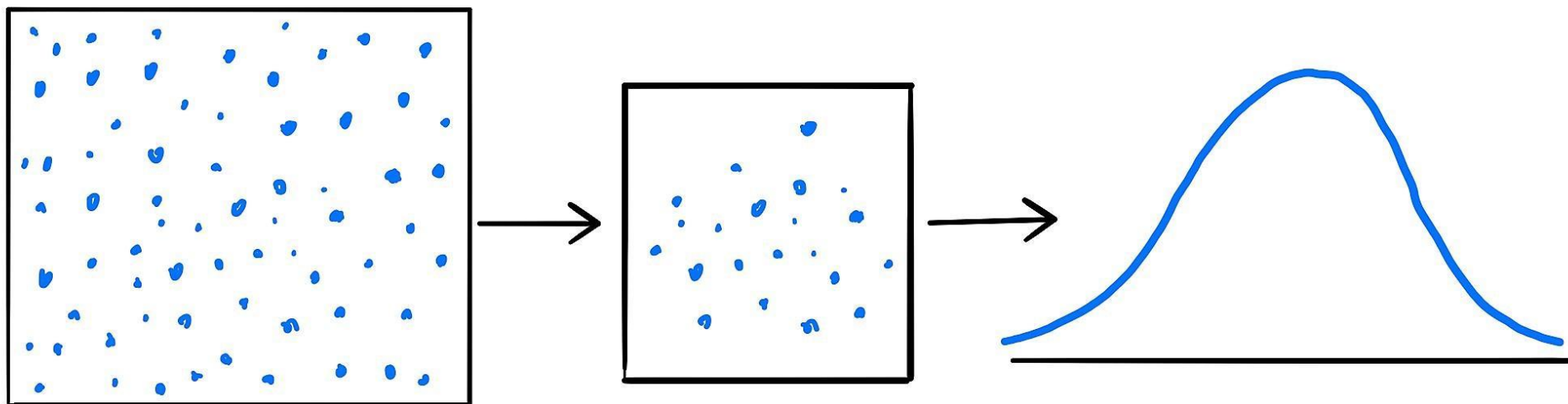


Machine Learning

Samatrix Consulting Pvt Ltd

Resampling Methods



Resampling Methods

- Resampling methods involve draw samples from a training set repeatedly.
- Then refit the model of interest on each sample. In this process, we obtain additional information about the fitted model.
- This information would not be available if we fit the model only once using the original training sample.
- The resampling methods involve fitting a machine learning method multiple times by using different subset of training data.
- Hence the resampling methods may be computationally expensive.
- However, due to recent advances in computing power, the resampling methods are no longer prohibitive.

What are resampling methods?

- Tools that involves repeatedly drawing samples from a training set and refitting a model of interest on each sample in order to obtain more information about the fitted model
 - Model Assessment: estimate test error rates
 - Model Selection: select the appropriate level of model flexibility
- They are computationally expensive! But these days we have powerful computers 😊
- Two resampling methods:
 - Cross Validation
 - Bootstrapping

Resampling Methods

- Basic steps
 - Repeatedly draw samples from a training set
 - Fit a model for the outcome of interest on each sample
 - Estimate information about the fitted model
- Example:
 - Goal: Estimate the variability of a linear regression fit
 - Draw different samples from the training data and fit OLS regression to each
 - Examining the extent to which the resulting fits differ.
- Can be useful in both model selection and model assessment

Resampling Methods

- The most commonly used resampling methods are **cross-validation** and **bootstrap**.
- For example, we can use cross-validation to estimate the error associated with a machine learning method to evaluate its performance or select the level of flexibility.
- The process of evaluating the performance of a model is called **model assessment**.
- Whereas the process of selecting the level of flexibility of a model is called the **model selection**.
- We use bootstrap methods to measure the accuracy of a given machine learning methods.

Cross Validation

- In the previous chapters, we discussed how the test error rates and the training error rates could be different from each other.
- If we do not have a very large test data set to estimate the test error rate, we can use different techniques to estimate the test error rate using the available training data.
- By using a class of methods, we can estimate the test error rates by holding out a subset of training observation from the fitting process and then applying the machine learning methods to the held out observations.



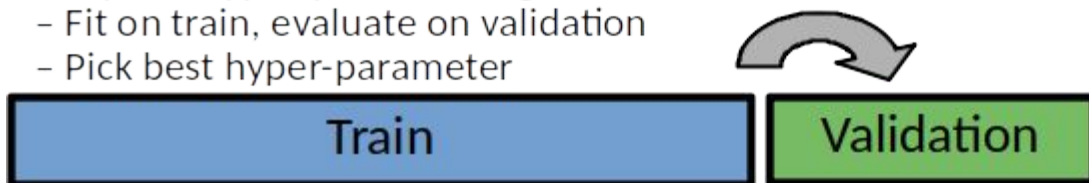
Model evaluation: hold-out a test set

- `mod.fit(X_train, y_train)`
- `mod.predic(X_test)`

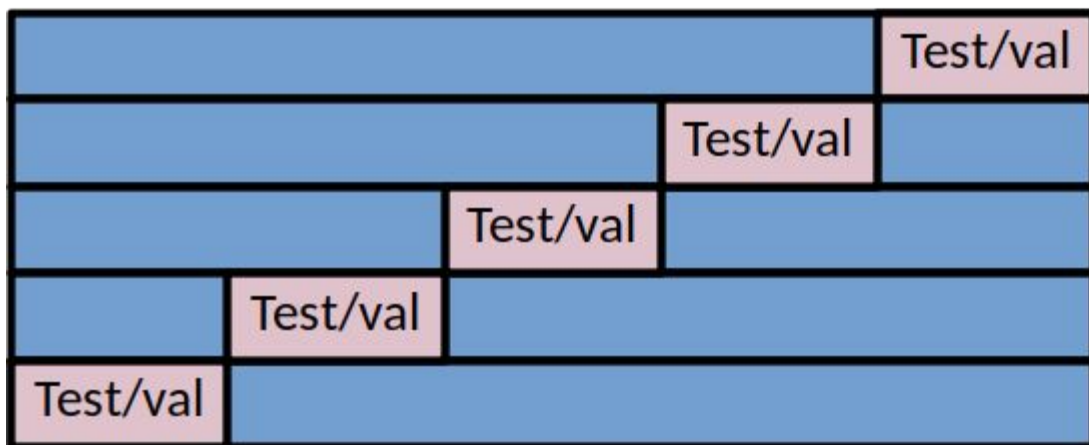


Model selection: hold-out a validation set

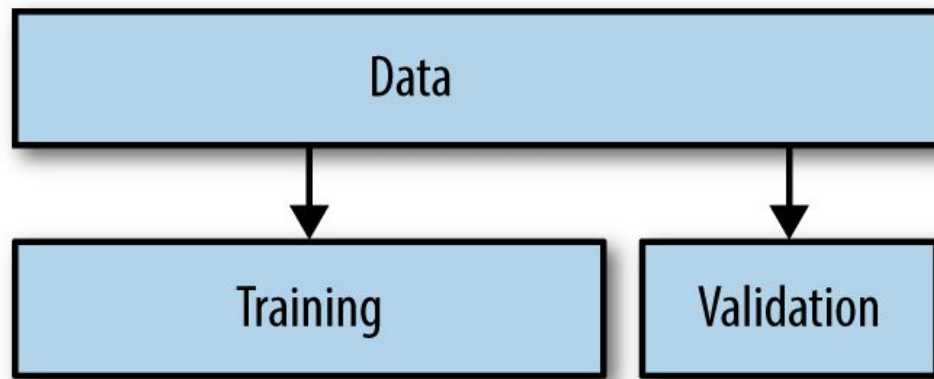
- Explore hyper-parameters grid
- Fit on train, evaluate on validation
- Pick best hyper-parameter



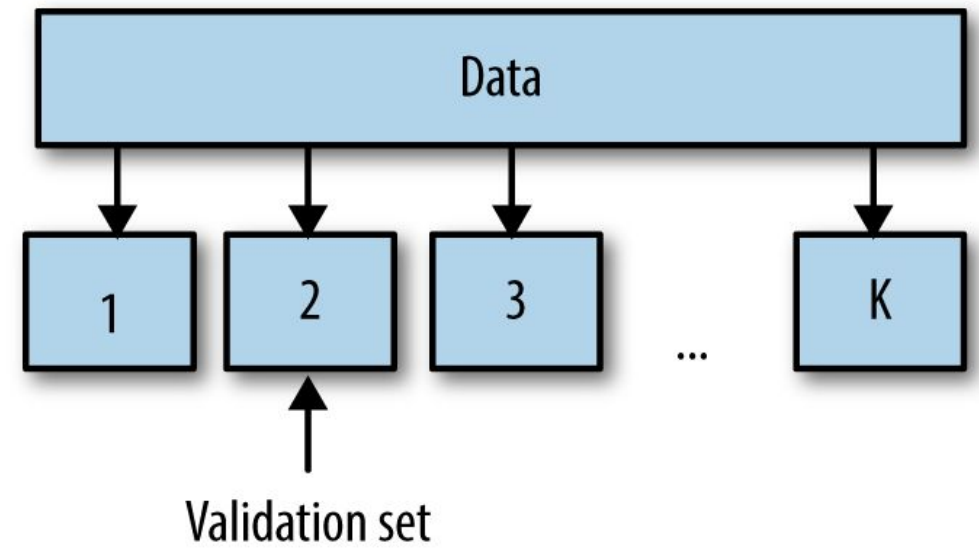
Cross-validation: when dataset is too small for to apply hold-out strategy. CV Can be used for model evaluation or/and model selection.



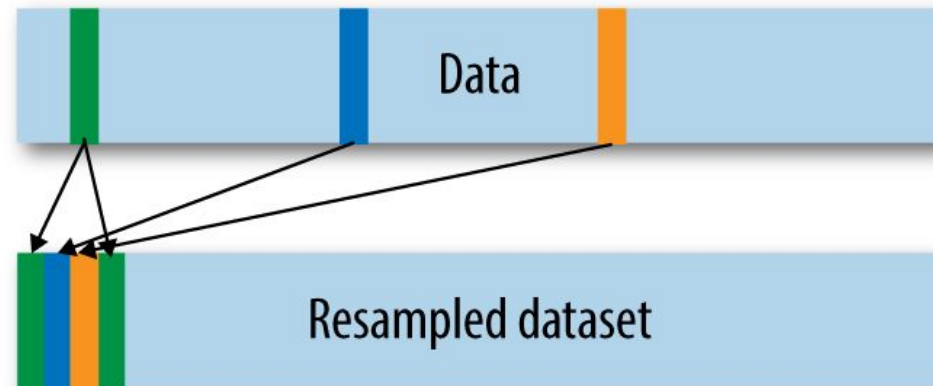
Hold-out validation



K-fold cross validation



Bootstrap resampling



<u>Resampling Method</u>	<u>Application</u>	<u>Sampling procedure used</u>
<u>Bootstrap</u>	Standard deviation, confidence interval, hypothesis testing, bias	Samples drawn at random, with replacement
<u>Jackknife</u>	Standard deviation, confidence interval, bias	Samples consist of full data set with one observation left out
<u>Permutation</u>	Hypothesis testing	Samples drawn at random, without replacement.
<u>Cross-validation</u>	Model validation	Data is randomly divided into two or more subsets, with results validated across sub-samples.

Validation Set Approach

- Validation set approach is used to estimate the test error rate.
- In this approach, we randomly divide the available set of observations into two parts: **training** set and **validation set or hold-out set**.
- We use the training set to fit the model and then use the fitted model to predict the responses for the observations in the validation set.
- We use the validation set error rate to estimate the test error rate.

Validation Set Approach



- We have demonstrated the schematic display of the validation set approach in the Figure 1.
- We have randomly split a set of n observations into a training set and a validation set.
- The training set is shown in blue and it contains observations, 7, 22, and 13 among others.
- The validation set is shown in beige and it contains observation 91 among others.
- We use a machine learning model to fit on the training set and use validation set to evaluate its performance.

Validation Set Approach

- Even though the validation set approach is conceptually simple and is easy to implement, it suffers two drawbacks:
 - The variance of the estimate of the test error rate using validation approach can be high because it depends how the observations have been split between the training set and the validation set
 - In the validation approach we use only a subset of the observations to fit the model. The machine learning models do not perform well on fewer observation and the validation set error rate may be an overestimation of the test error rate for the model that has been fit on the entire data set.

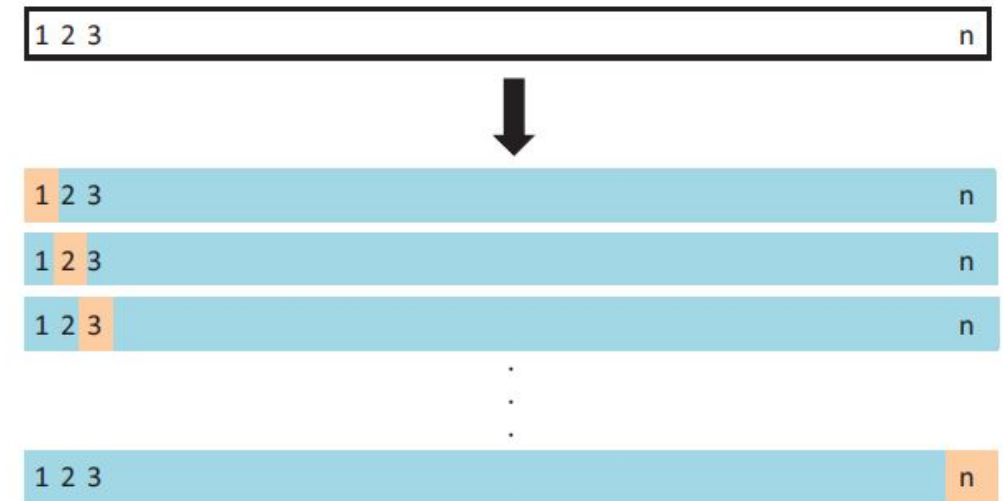
Leave-one-out Cross-Validation

- Leave-One-Out Cross Validation (LOOCV) is similar to the validation set approach but it tries to address the drawbacks of validation approach.
- Similar to the validation set approach, the LOOCV also split the data set into two parts.
- Instead of creating two subsets of comparable size, we use only one observation (x_1, y_1) for the validation set and use the remaining $(n - 1)$ observations $\{(x_2, y_2), \dots, (x_n, y_n)\}$ for the training set.
- We fit a machine learning model on the $(n - 1)$ training observations, make prediction $\hat{y}_1$ using the excluded observation x_1 , and calculate the $MSE_1 = (y_1 - \hat{y}_1)^2$.
- But the MSE_1 is highly variable because it is based on a single observation.

Leave-one-out Cross-Validation

- Now we repeat the procedure by selecting (x_2, y_2) for validation data, training machine learning model on $n - 1$ observation
- $\{(x_1, y_1), (x_3, y_3) \dots, (x_n, y_n)\}$, and computing $MSE_2 = (y_2 - \hat{y}_2)^2$.
- If we repeat this approach n times, we get n squared errors $MSE_1, \dots, MSE_n$.
- The LOOCV estimate for the test MSE can be calculated by taking the average of all these n test error estimates:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n MSE_i$$



Leave-one-out Cross-Validation

- LOOCV has two major advantages over the validation set approach.
- Firstly, LOOCV has far less bias as compared to validation set approach.
- In the case of LOOCV, we repeatedly fit the machine learning model on the $n - 1$ training observations (almost as many as are in the entire data set), whereas in the case of validation set approach, we split the training set into two comparable sizes.
- Hence the LOOCV approach does not overestimate the test error rate as much as the validation set approach tends to overestimate.

Leave-one-out Cross-Validation

- Secondly, due to randomness in the training/validation set splits, the validation set approach will yield different results when applied repeatedly.
- While if we perform LOOCV multiple times, we will always get the same results because there is no randomness in the training/validation set splits.
- On the other hand, LOOCV is computationally expensive because the model has to be fit n times.
- If n is large and each individual model is slow to fit, this can be very time consuming.

Leave-one-out Cross-Validation

- LOOCV is a general method. We can use it for any kind of predictive modeling.
- We can also use it with logistic regression or linear discriminant analysis

k-fold Cross Validation

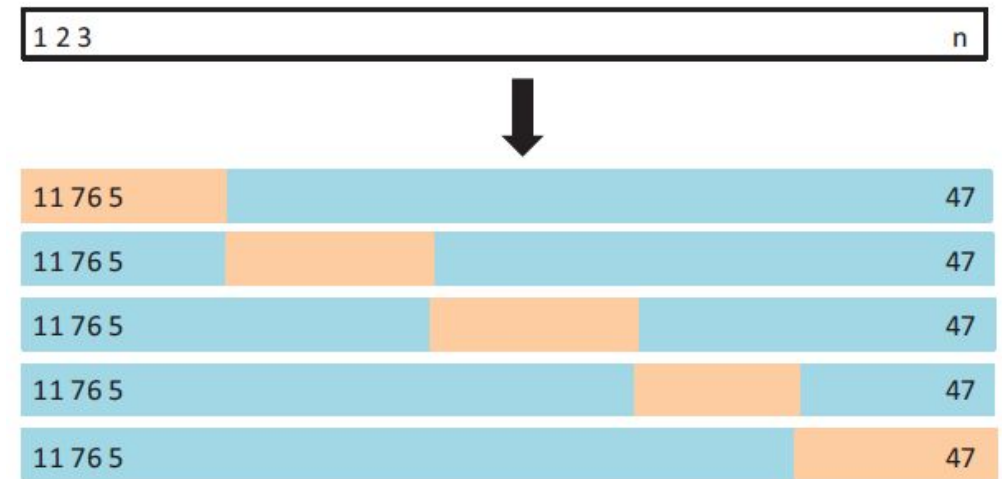
- k -fold cross validation is an alternative to LOOCV. In this method, we randomly divide the data set into k groups or folds of approximately equal size.
- We treat the first fold as a validation set and a machine learning method is fit on the remaining $k - 1$ folds.
- We calculate the mean squared error, MSE_1 , on the observations in the held-out fold.
- We repeat the procedure k times by treating a different group of observations as a validation set.

k-fold Cross Validation

- Thus, we get k estimates of the test error $MSE_1, MSE_2, \dots, MSE_k$. Finally, we compute k -fold CV estimate by averaging these values

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$$

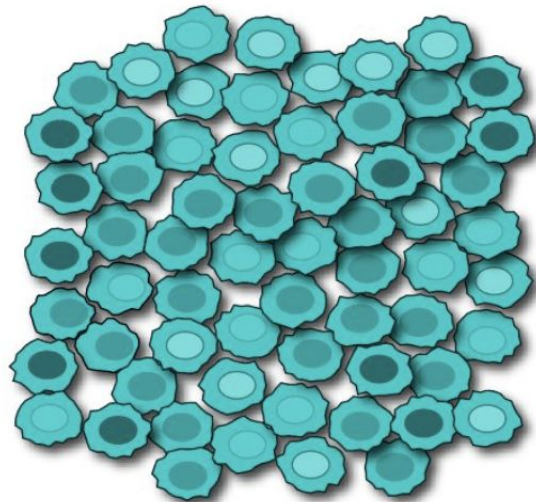
- A schematic of the k -fold cross validation approach has been demonstrated in Figure – 3.



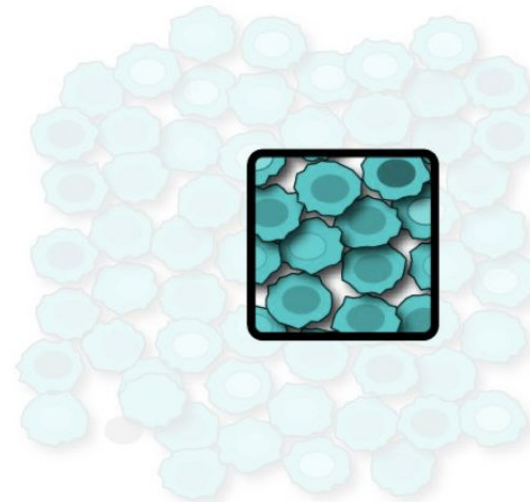
k-fold Cross Validation

- We can see that LOOCV is a special case of k -fold CV in which $k = n$.
- In practice, we typically perform k -fold CV using $k = 5$ or $k = 10$.
- The question arises about the advantage of using $k = 5$ or $k = 10$ instead of $k = n$.
- One of the major advantages is computational.
- The LOOCV requires fitting the learning method n times, which makes LOOCV computationally expensive when compared to the k -fold CV using $k = 5$ or $k = 10$.
- For example, if we take $k = 10$, the 10-fold CV requires fitting the learning method only 10 times which may be much more feasible.

Population



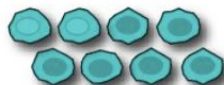
Selection



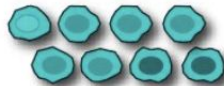
Experimental
Sample



n=1



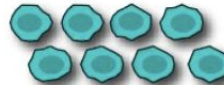
n=2



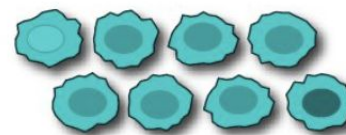
n=.....



n=1000



Resample
(bootstrap)



n=1



population
inferences



population
inferences

The Bootstrap

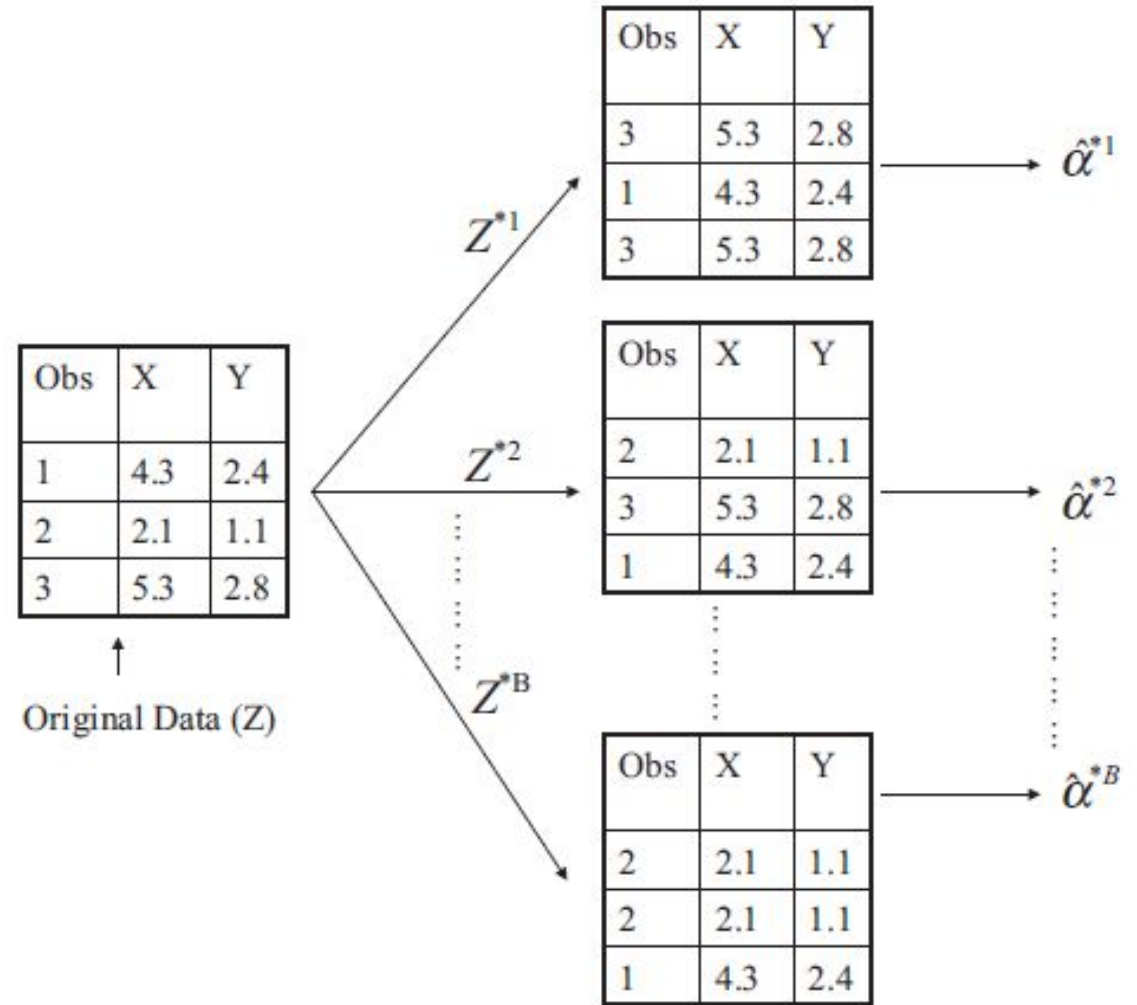
- The bootstrap is widely acceptable and very powerful tool.
- It is used to quantify the uncertainty that are associated with a given machine learning method.
- For example, we can use the bootstrap to estimate the standard errors of the coefficients from a linear regression fit.
- It may not be useful for linear regression fit because standard statistical software such as R and Python automatically provide standard error.
- For a wide range of machine learning methods, the standard errors are difficult to obtain. We can use the bootstrap for such methods.

The Bootstrap

- The bootstrap estimate quantities about a population by averaging estimates from small data samples.
- The samples are constructed by drawing observations from a large data sample one at a time and returning to the sample after they have been chosen.
- This allows a given observation to be included in a given sample more than once.
- This approach of sample is known as **sampling with replacement**.

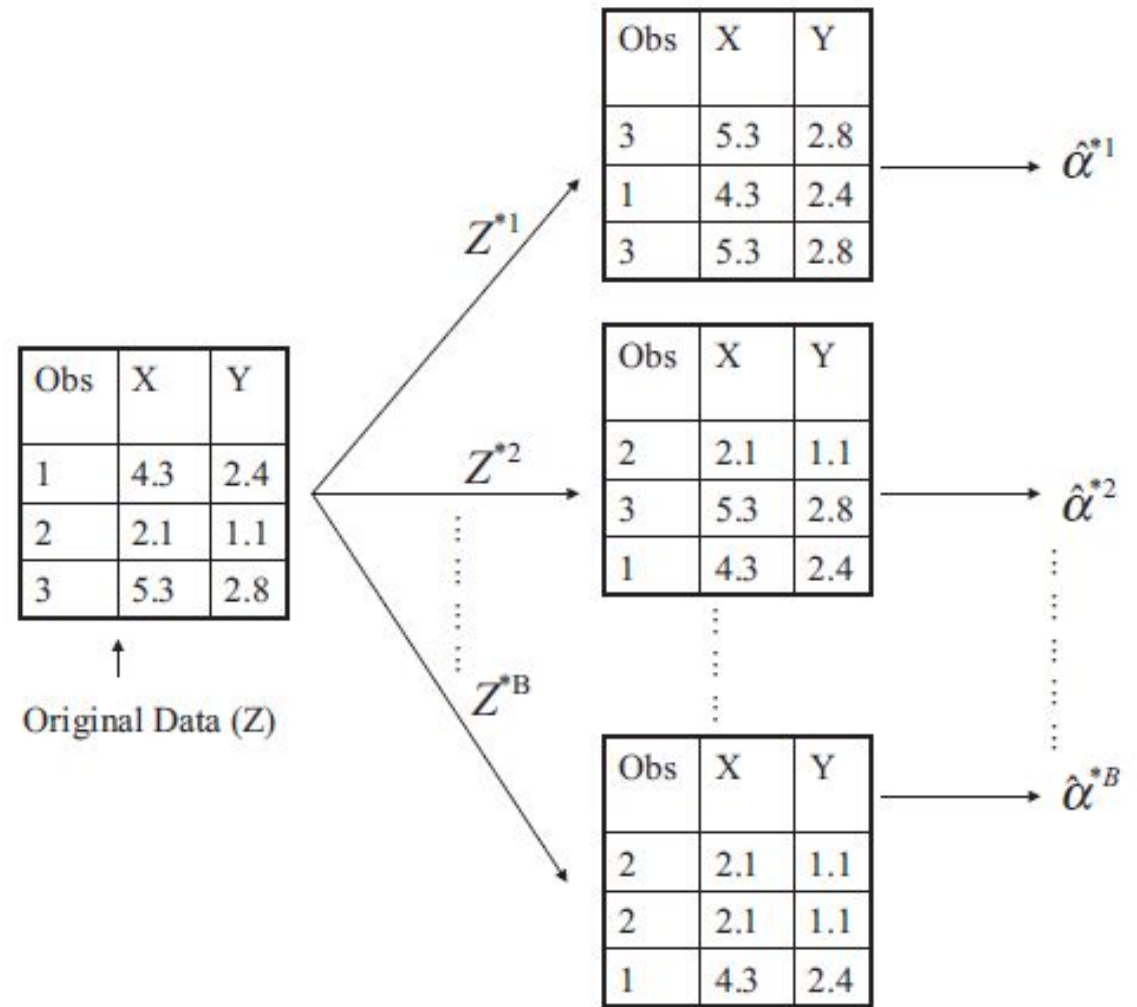
The Bootstrap

- In this example, we have a simple data set which we can call Z and it contains only $n = 3$ observations.
- From this dataset, we randomly select n observations to produce a bootstrap dataset Z^{*1} .
- We performed this sampling by replacement due to which same observation can be taken in the bootstrap dataset more than once.
- In the current example, for Z^{*1} dataset, we have the first observation once and the third observation twice but the second observation is not present in the dataset.



The Bootstrap

- We can use the Z^{*1} to calculate the bootstrap estimate for α , which we can call $\hat{\alpha}^{*1}$.
- We repeat this procedure B times by keeping a large value of B .
- Thus, we produce B different bootstrap data sets, $Z^{*1}, Z^{*2}, \dots, Z^{*B}$.
- We also produce corresponding B α estimates, $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \dots, \hat{\alpha}^{*B}$.
- We can now compute the standard error of these bootstrap estimates.



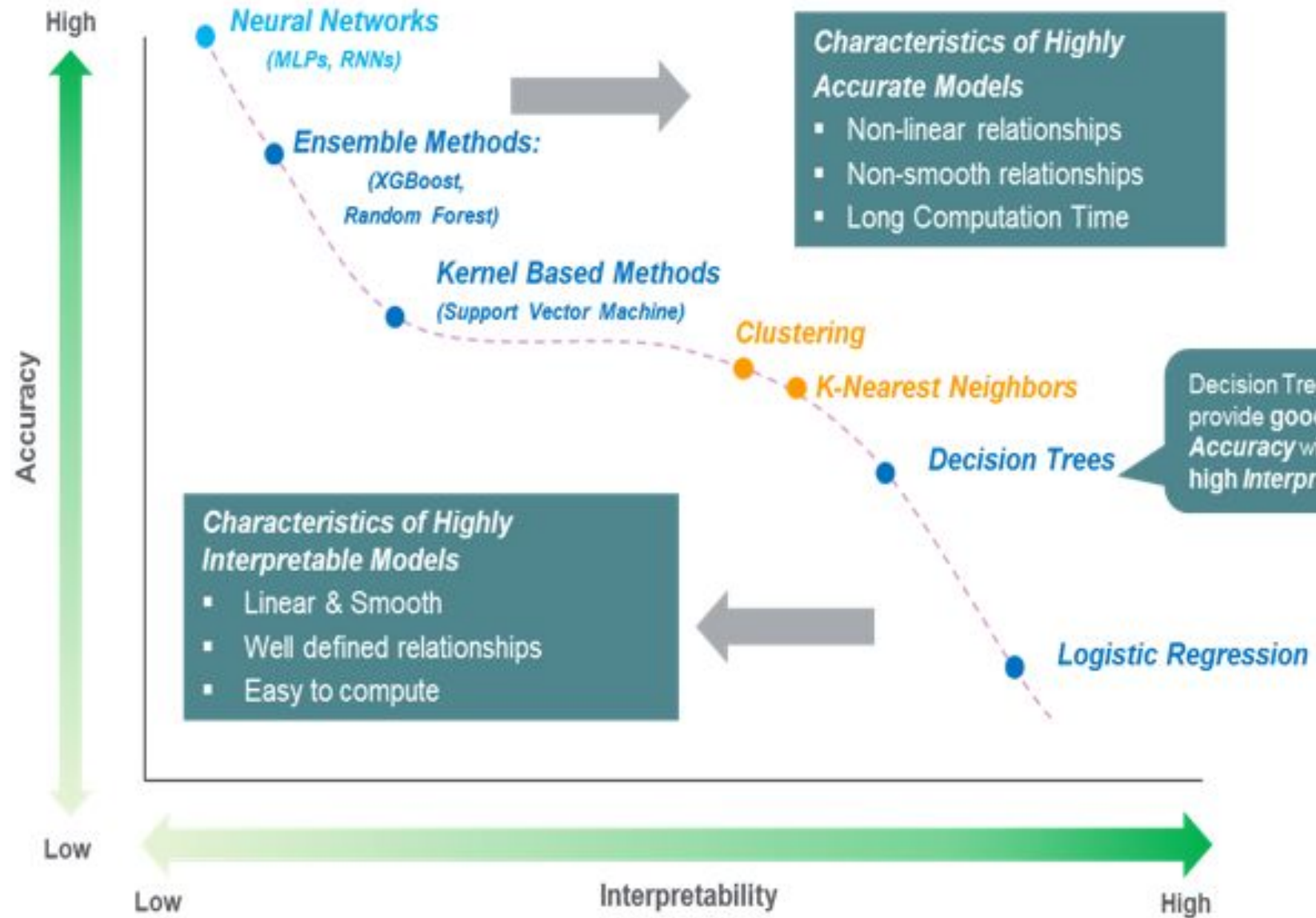
Linear Model Selection

Linear Model Selection

- In previous chapters, we discussed about the standard linear model in the regression setting

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

- In the following chapters, we will study about the non-linear models. Before moving to other models, we will study about the ways in which the linear model can be improved using alternative fitting procedures.
- These alternative fitting methods can help achieve better **prediction accuracy** and **model interpretability**.



Prediction Accuracy

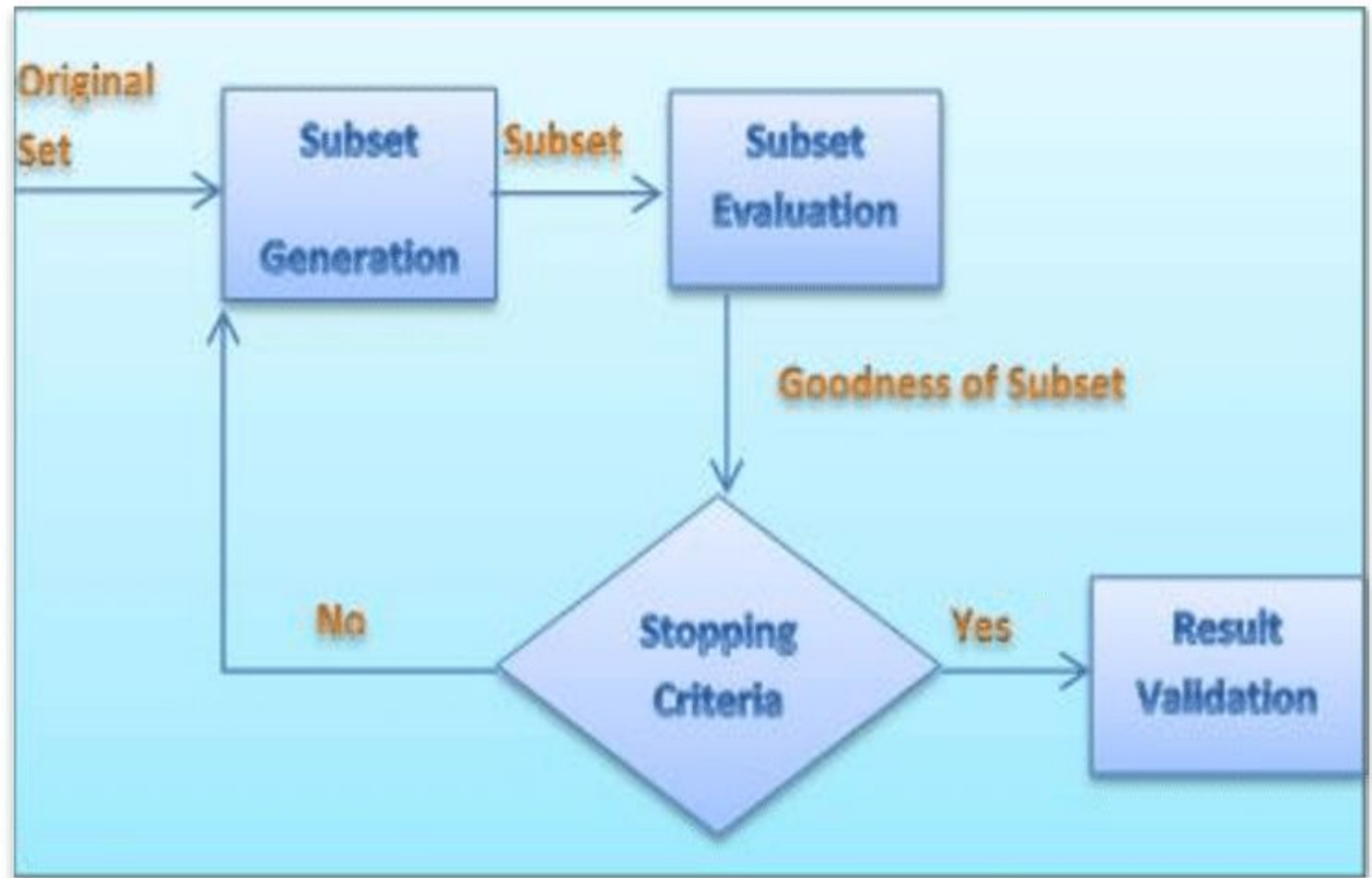
- If the true relationship between the response and the predictor is approximately linear, the least square estimates will have low bias.
- If $n \gg p$, which means that number of observations n is much larger than p , the number of variables, the least square estimates have low variance too.
 - They will perform well on test observations.
- However, if n is not much larger p , there might be a lot of variability in the least square fit.
 - This results in overfitting and poor results on test observations.
- If $n < p$, then we cannot find unique least square coefficient estimates.
 - In this case the variance is infinite and we cannot use the method at all.
- We can use constraining or shrinking the estimated coefficients to reduce the variance at the cost of negligible increase in bias. This leads to substantial improvement in the accuracy.

Model Interpretability

- On several occasions, some or many variables that we use in a multiple regression model are not associated with the response.
- We can remove these variables by setting the corresponding coefficient estimates to zero.
- This results in a model that is easier to interpret.
- We can use feature selection or variable selection methods to exclude irrelevant variables.
- In this section, we will study about three important class of methods: Subset selection, Shrinkage, Dimension Reduction.

Subset Selection

- In this approach, we identify a subset of p predictors that are related to the response.
- This includes best subset and stepwise model selection procedures.



Best Subset Selection: Example with 3 Variables



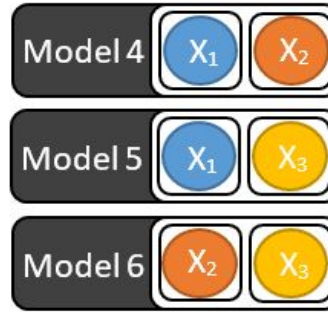
Step 1: Consider All Possible Models

By listing all possible combination of variables

Models with 1 variable:



Models with 2 variables:



Models with 3 variables:



Step 2: Identify the Best Model of Each Size

By choosing the one with the lowest sum of squared errors or the highest R^2

Best model with 1 variable:



Best model with 2 variables:



Best model with 3 variables:



Step 3: Identify the Best Overall Model

By choosing the one with the lowest AIC (or BIC) or the highest adjusted R^2

Best overall model:



Best Subset Selection

- In this method, we fit the least square regression for each possible combination of the p predictors separately.
- We fit all p models that contain exactly one predictor, all $\binom{p}{2} = p(p-1)/2$ models that contain exactly two predictors, and so forth.
- Finally, we look at all of the resulting models to identify the best one.

Best Subset Selection

- The process of finding the best model among 2^p possibilities is not easy. So, we adopt the two-stage algorithm as follows
 - Let M_0 denote the null model. It contains no predictors. Hence in this case, the we can predict the sample mean of each observation using the model.
 - For $k = 1, 2, \dots p$:
 - Fit all $\binom{p}{k}$ models that contain exactly k predictors
 - Pick the model which has the smallest RSS or largest R^2 among these $\binom{p}{k}$ models and name it M_k
 - Select a single best model from among $M_0, \dots, M_p$ by using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2

Best Subset Selection

- In the step 2, we identify the best model for each subset size.
- This helps reduce the problem from identifying the best model from 2^p possible models to one of $p + 1$ possible models.
- This is a critical task because as the number of the features that we include in the model increases, RSS decreases and R^2 increases.
- Hence, we may end up with a model that involves all the variables.
- Low RSS or a high R^2 are associated with a low training error whereas we are interested in low test error.
- Therefore, in step 3, we used the cross-validated prediction error, C_p , BIC, or adjusted R^2 in order to select among $M_0, \dots, M_p$

Best Subset Selection

- While the best subset selection is simple, it has computational limitation.
- The number of possible models that we should consider, grows rapidly as p increases.
- Because for p predictors, there are 2^p subsets.
- For $p = 10$, we have to consider approximately 1000 possible models.
- For $p = 20$, we have approximately one million possibilities.
- The best subset selection model is computationally infeasible for $p > 40$ even with fast modern computers.
- So, in next section we study about the computationally efficient alternatives to best subset selection.

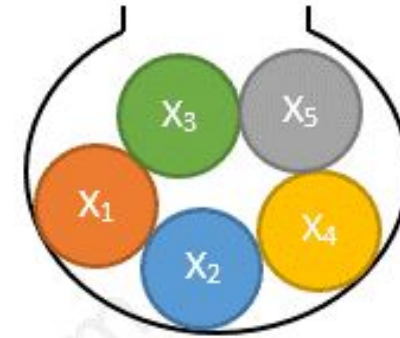
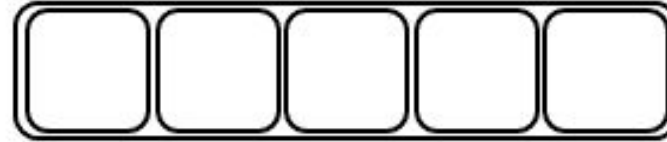
Stepwise Selection

- As we have seen that the best subset selection cannot be used where $p > 40$ due to computational reasons.
- Due to large p , the best subset selection may also suffer from statistical problems.
- If the search space is large, we may end up in finding in a model that looks good on the training data but may not be an efficient model for unseen data.
- Hence, we need a far more restrictive set of models.

Forward stepwise selection example with 5 variables:

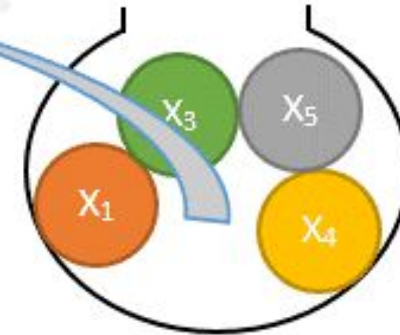
Start with a model with no variables

Null Model



Add the most significant variable

Model with 1 variable



Keep adding the most significant variable until reaching the stopping rule or running out of variables

Model with 2 variables



Forward Stepwise Selection

- The forward stepwise selection begins with a model without any predictors.
- It then adds the predictors to the model, one-at-a-time, until all the predictors are added to the model.
- At each step, the predictor that gives the greatest additional improvement to the fit is added to the model.
- The procedure is as follows
 - Let M_0 denote the null model. It contains no predictors.
 - For $k = 0, \dots, p - 1$:
 - Take all $p - k$ models that can add one additional predictor to the existing predictors in M_k
 - Pick the best among the $p - k$ models which has the smallest RSS or largest R^2 and name it M_{k+1}
 - Select a single best model from among $M_0, \dots, M_p$ by using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2

Forward Stepwise Selection

- Even though the forward stepwise selection has a computation advantage over the best subset selection, it cannot always find the best model out of all 2^p models containing subsets of the p predictors.
- For example, if we have a data set with $p = 3$ predictors.
- The best possible one-variable model contains X_1 whereas the best possible two-variable model contains X_2 and X_3 .
- In this case the forward stepwise selection will not be able to select the best possible two-variable model because M_1 will contain X_1 , so M_2 must also contain X_1 along with an additional variable.

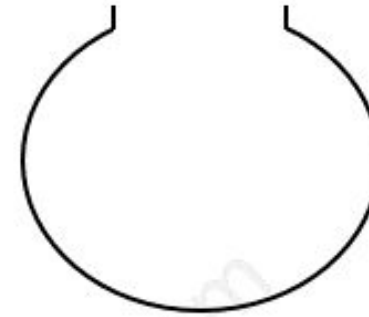
Forward Stepwise Selection

- We can use the forward stepwise selection for high-dimensional setting where $n < p$.
- We can construct sub models $M_0, \dots, M_{n-1}$ only because each sub model will fit using least square.
- If $p \geq n$, these models will not give unique results.

Backward stepwise selection example with 5 variables:

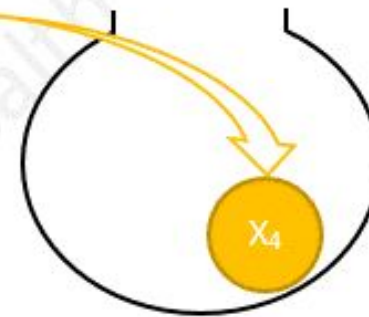
Start with a model that contains all the variables

Full Model



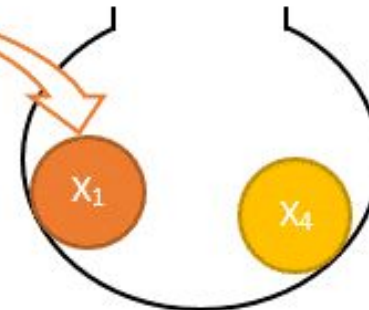
Remove the least significant variable

Model with 4 variables



Keep removing the least significant variable until reaching the stopping rule or running out of variables

Model with 3 variables



Backward Stepwise Selection

- Backward Stepwise Selection is also an efficient alternative to best subset selection.
- In contrast to the Forward stepwise selection, it starts with the full least square model with all p predictors and then iteratively removes the least useful predictor one-at-a-time.
- The steps are as follows.
 - Let M_p denote the full model. It contains all p predictors.
 - For $k = p, p - 1, \dots, 1$:
 - Take all k models that contain all the predictors except one of the predictors in M_k , for a total of $k - 1$ predictors
 - Pick the best among the k models which has the smallest RSS or largest R^2 and name it M_{k-1}
 - Select a single best model from among $M_0, \dots, M_p$ by using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2

Backward Stepwise Selection

- For the backward stepwise selection, the number of samples n should be larger than the number of variables p , so that full model can be fit.
- If $n < p$ and p is very large, the forward stepwise selection is the only viable subset method.

Cp, AIC, and BIC

- While fitting a model to a training data using least squares, we estimate the regression coefficients by minimizing training MSE ($MSE = RSS/n$).
- Due to this the training MSE is an underestimate of test MSE .
- Moreover, by increasing the variables in the model, we may reduce the training error but the test error will not reduce.
- Hence, we cannot use training set RSS and training set R^2 for subset selection

Cp, AIC, and BIC

- We can use several techniques for adjusting the training error for the model size.
- We can use these approaches to select among a set of models with different number of variables.
- Four such approaches are: C_p , Akaike information criterion (AIC), Bayesian information criterion (BIC) and adjusted R^2 .

Cp, AIC, and BIC

- C_p : For a least square model with d predictors, C_p estimate of test MSE can be given by the following equation

$$C_p = \frac{1}{n} (RSS + 2d\hat{\sigma}^2)$$

- In the above equation $\hat{\sigma}^2$ provides an estimate of the variance of the error ϵ .
- In order to adjust the underestimation of test error by the training error, C_p adds a penalty of $2d\hat{\sigma}^2$ to the training RSS .
- It is clear that the penalty increases with the increase in the number of predictors in the model.

Cp, AIC, and BIC

- **AIC:** For a large class of models fit, the AIC criterion can be defined by maximum likelihood.
- For a model with Gaussian errors, the maximum likelihood and least square are same.
- Hence, we can represent AIC by

$$AIC = \frac{1}{n\hat{\sigma}^2} (RSS + 2d\hat{\sigma}^2)$$

- In this case for simplicity, we have omitted an additive constant.
- For least square models C_p and AIC are proportional to each other.

Cp, AIC, and BIC

- BIC: We can derive BIC using Bayesian point of view.
- However, it looks similar to C_p and AIC.
- For a least square model with d predictors, the BIC can be given by

$$BIC = \frac{1}{n}(RSS + \log(n)d\hat{\sigma}^2)$$

- We select the model with lowest BIC value.
- BIC results in selection of smaller models than C_p .

Shrinkage Method

- We have studied that subset selection methods results into a linear model that uses least squares and contains a subset of the predictors.
- Alternatively, we can find a model that contains all the p predictors and uses a technique to **regularize** the coefficient estimates, or equivalently, shrinks the coefficient estimates towards zero.
- By shrinking the coefficient estimates we can significantly reduce their variance.
- The two best-known techniques for shrinking the regression coefficients towards zero are **ridge regression** and the **lasso**.

Ridge Regression

- We have already studied that the least square fitting procedure estimates $\beta_0, \beta_1, \dots, \beta_p$ using the values that minimize

$$RSS = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

- The Ridge regression is very similar to the least square.
- The difference is that we estimate the coefficients by minimizing a slightly different quantity.

Ridge Regression

- We can say that the ridge regression coefficient estimates $\tilde{\beta}^R$ are that values that minimizes

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = RSS + \lambda \sum_{j=1}^p \beta_j^2$$

- In this case $\lambda \geq 0$ is a **tuning parameter** that we can determine separately. The equation trades off different criteria.
- Similar to the least squares, the ridge regression tries to find the coefficients estimates that can fit the data well by making the RSS small.
- But the second term $\lambda \sum_{j=1}^p \beta_j^2$ is called **shrinkage penalty**.
- The shrinkage penalty is small when $\beta_1, \dots, \beta_p$ are close to zero.

Ridge Regression

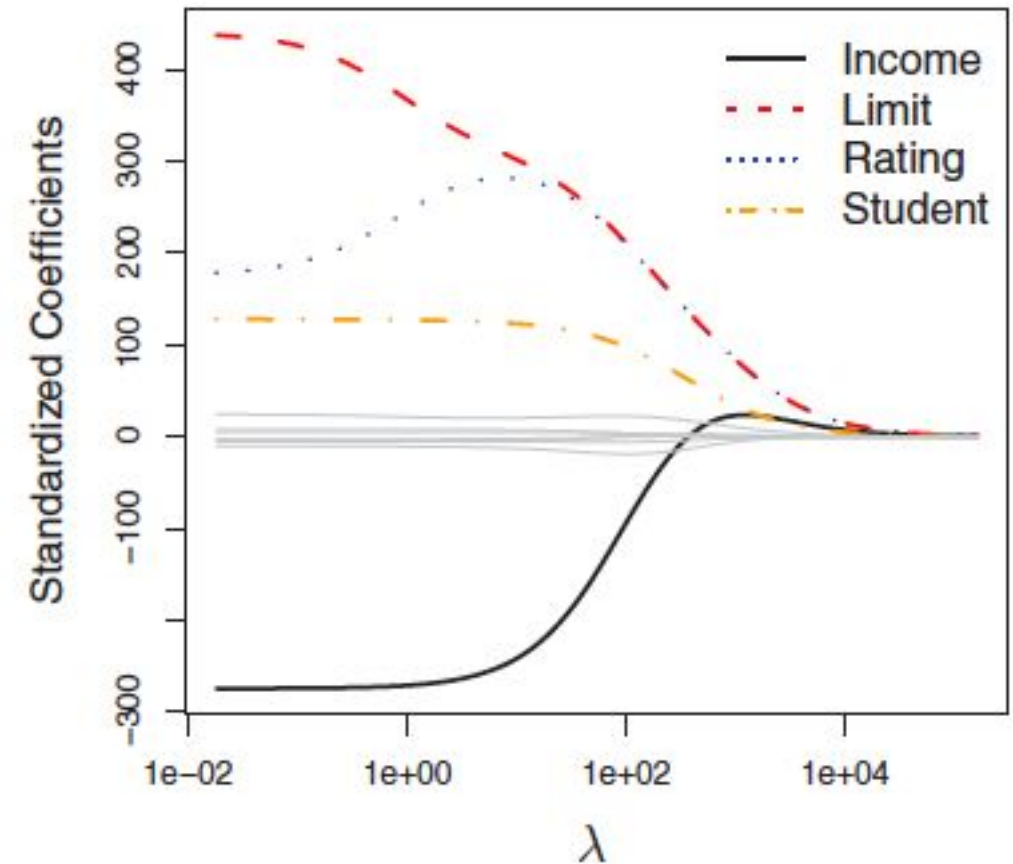
- So, it tries to shrink the estimates β_j towards zero.
- The tuning parameter λ controls the relative impact of these two terms on the regression coefficient estimates.
- When $\lambda = 0$, there is no effect of the penalty effect and the ridge regression will produce the least square estimates.
- When the $\lambda \rightarrow \infty$, the impact of the shrinkage penalty grows.
- In this case the ridge regression coefficient estimates will approach zero.
- In contrast to the least squares, the ridge regression will produce a different set of coefficient estimates $\hat{\beta}_\lambda^R$, for each value of λ .

Ridge Regression

- You should also note that the shrinkage penalty is applied to $\beta_1, \dots, \beta_p$, but not to the intercept β_0 because the intercept is a measure of the mean value of the response.
- The ridge regression uses l_2 (pronounced “ell 2”) penalty.
- The l_2 norm of a coefficient vector β is given by $||\beta||_2 = \sqrt{\sum \beta_j^2}$

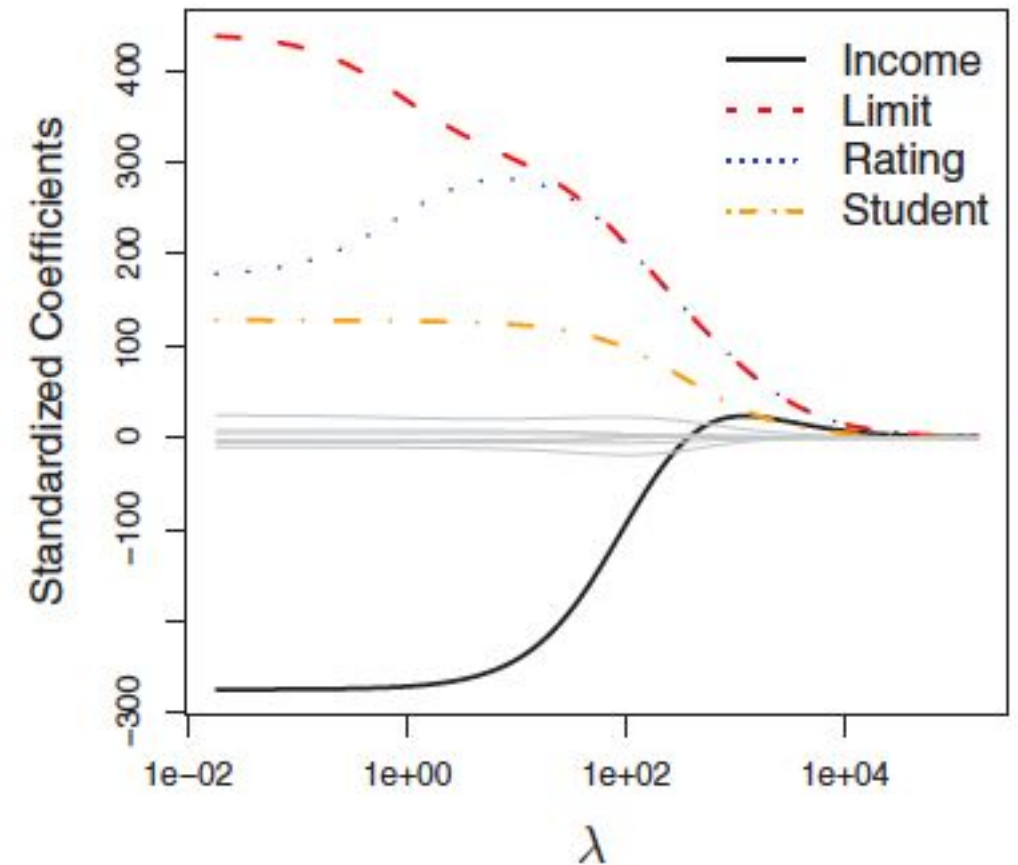
Credit Card Default Case Study

- In the figure – 5, we have demonstrated the ridge regression coefficient estimates for the credit card default data.
- Each curve corresponds to the ridge regression coefficient estimate for one of the ten variables that have been plotted as a function of λ .
- The black solid line demonstrates the regression coefficient for the income coefficient with respect to λ .
- At the extreme left-hand side of the plot, the value of λ is zero.



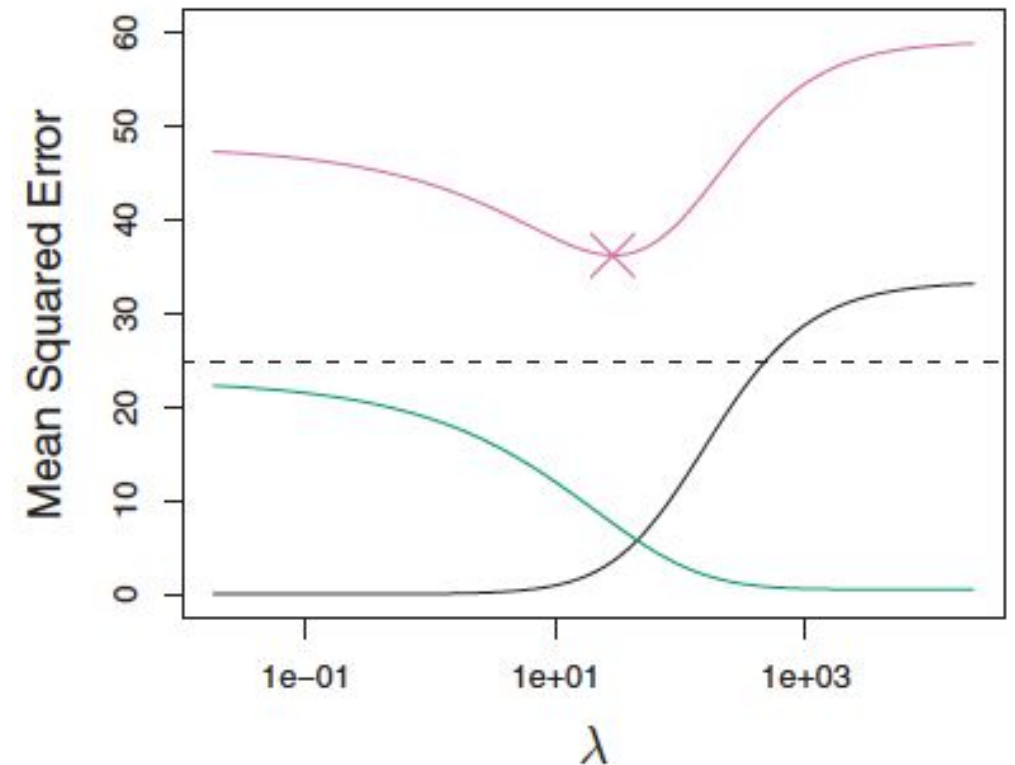
Credit Card Default Case Study

- So, the corresponding ridge regression coefficients and least square estimates are the same.
- As the value of λ increases, the ridge regression estimates shrink towards zero.
- For an extremely large value of λ , all the ridge coefficient estimates are zero.
- This represents a null model which does not contain any predictors.



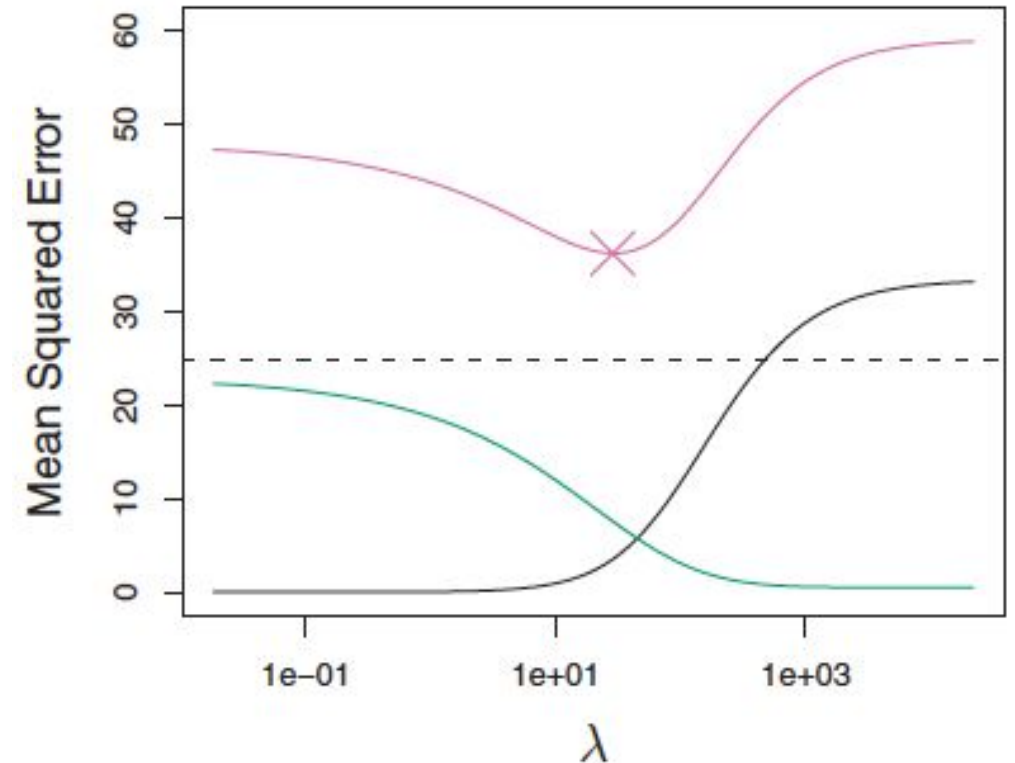
Why Ridge Regression Improves Over LS

- The improvement in Ridge Regression is due to bias-variance trade off.
- As λ increase, the flexibility of the ridge regression fit decreases.
- This leads to a decrease in variance and an increase in bias.
- We have illustrated the same in Figure – 6.
- We have plotted the plot using a data set which contains $p = 45$ predictors and $n = 50$ observations.
- The green curve in the plot shows the variance of the ridge regression predictions as a function of λ .
- At $\lambda = 0$, the variance is high but there is no bias



Why Ridge Regression Improves Over LS

- As we increase λ , we observe the substantial reduction in the variance of the predictors and slight increase in bias.
- This is due to the shrinkage of ridge coefficient estimates.
- Until $\lambda = 10$, the variance decreases rapidly but the bias (plotted in black) increases slowly.
- The MSE, which is function of variance plus squared bias, drops considerably as the value of λ increases from 0 to 10.
- Beyond this point the variance decreases slowly whereas the bias increases rapidly with increase in the value of λ .
- In this case we achieved minimum MSE approximately at $\lambda = 30$.



When to use Ridge Regression

- We should use the Ridge Regression when LSE has high variance.
- In the situation when the relationship between the response and the predictors is close to linear, the least square estimates will have low bias but high variance.
- When the number of predictors is close to number of observations, the least square estimates will be extremely variable.
- If $p > n$, we will not have a unique solution for least square estimates.
- In this situation also, Ridge regression can perform very well by trading off a small increase in bias for a large decrease in variance.
- Ridge regression also has substantial computational advantages over the best subset selection, which requires searching through 2^p models.

Disadvantage of Ridge Regression

- Ridge regression shrinks all the coefficients towards zero but it will not set any of them to zero unless $\lambda = \infty$.
- Hence the final model of the Ridge Regression will include all p predictors in the final model.
- This will not create any issues with prediction accuracy, but it can create a challenge in the model interpretation in case the number of variables p is quite large.

The Lasso (L1 Regularization)

- The lasso is an alternative to the ridge regression that can overcome this disadvantage. The lasso coefficients $\hat{\beta}_\lambda^L$ minimize the quantity

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = RSS + \lambda \sum_{j=1}^p |\beta_j|$$

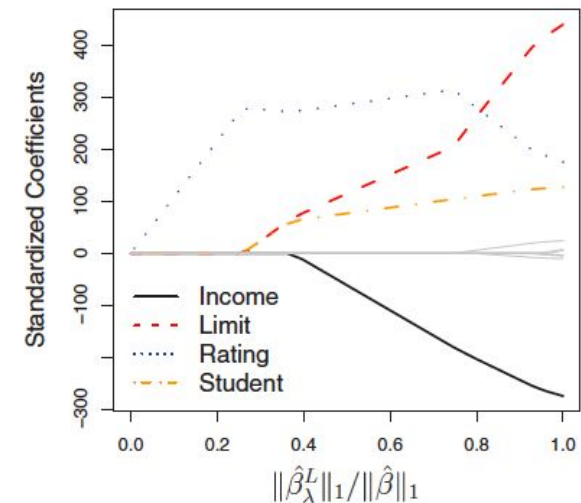
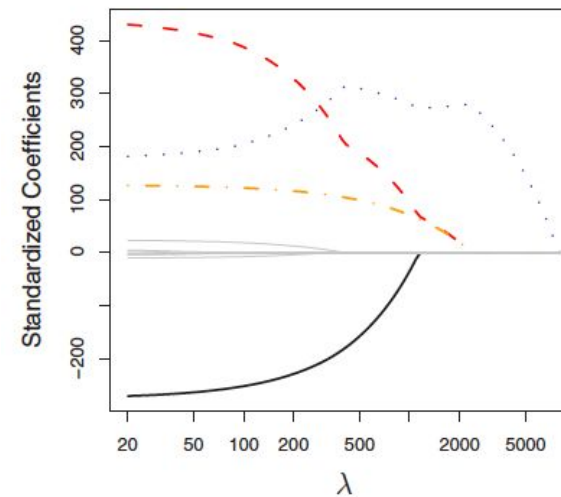
- The formulation of lasso and ridge regression is same.
- The only difference is the β_j^2 term in ridge regression penalty and $|\beta_j|$ term in the lasso penalty.
- The lasso uses l_1 (pronounced “ell 1”) penalty.
- The l_1 norm of a coefficient vector β is given by $\|\beta\|_1 = \sum |\beta_j|$

The Lasso (L1 Regularization)

- Similar to the ridge regression, the lasso also shrinks the coefficient estimates towards zero.
- But in the case of lasso, the l_1 penalty forces some of the coefficient estimates to exactly zero when the tuning parameter λ is sufficiently large.
- Hence the lasso performs the variable selection due to which the models generated from lasso are much easier to interpret than the ones produced by the ridge regression.
- We can also say that the lasso yields sparse model which means that the model involves only a subset of the variables.
- As in the case of the ridge regression, it is critical to select a good value of λ

Credit Card Default Case Study

- We have applied lasso to the credit card default case study.
- The results have been displayed in the Figure – 7.
- When $\lambda = 0$ the lasso returns the least square fit.
- For a significantly large value of λ , we get a null model where all the coefficient estimates are equal to zero.
- In between these two extremes, the lasso model and ridge model are quite different from each other.
- When we move from left to right in the right-hand panel, we notice that at first, the lasso model contains only the rating predictor.
- Then student and limit enter the model simultaneously followed by income.
- Eventually remaining variables enter the model.
- Depending on the value of λ , the lasso model can involve any number of variables.



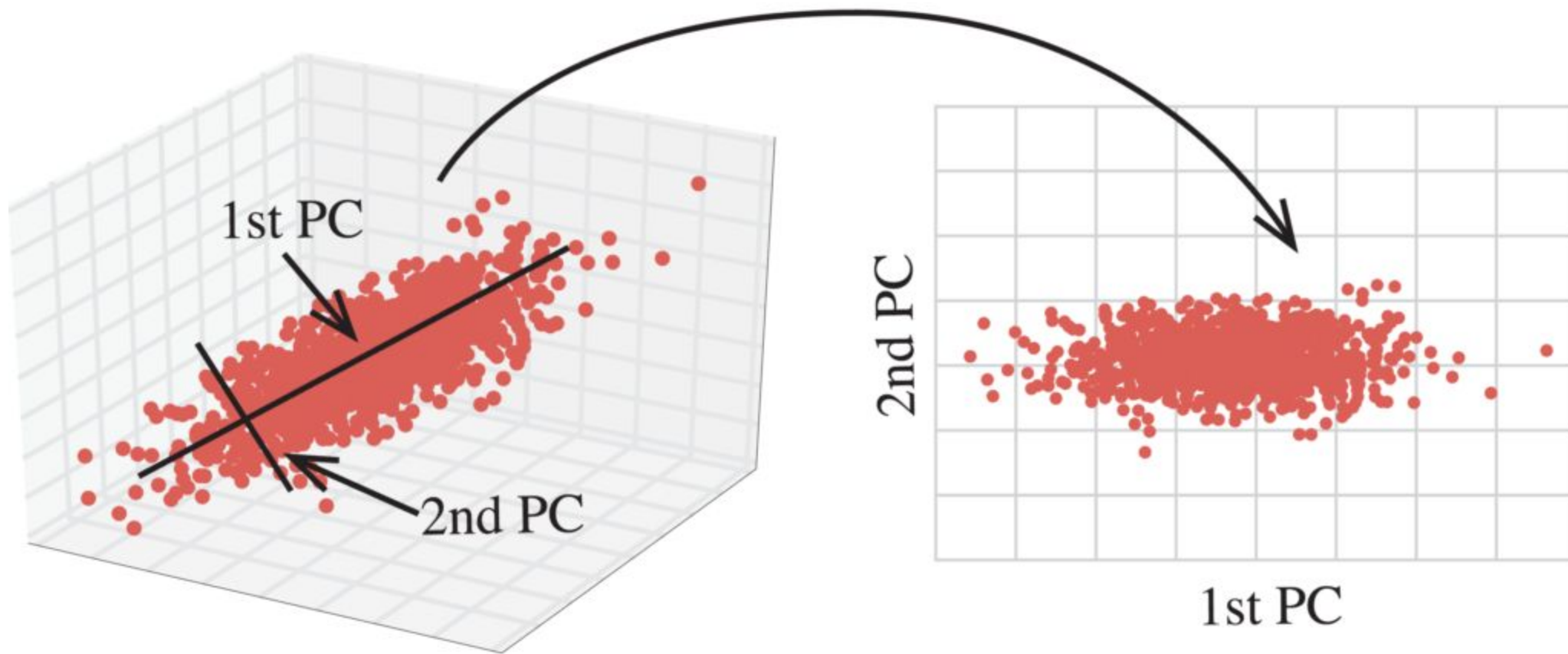
Selecting Tuning Parameter

- Implementation of ridge regression and the lasso regression requires a method of selecting a value of the tuning parameter λ .
- We can handle this problem using the cross-validation.
- We choose a grid of λ values and compute the cross-validation error for each value of λ .
- We select the tuning parameter for which cross-validation error is the smallest.

Dimension Reduction Methods

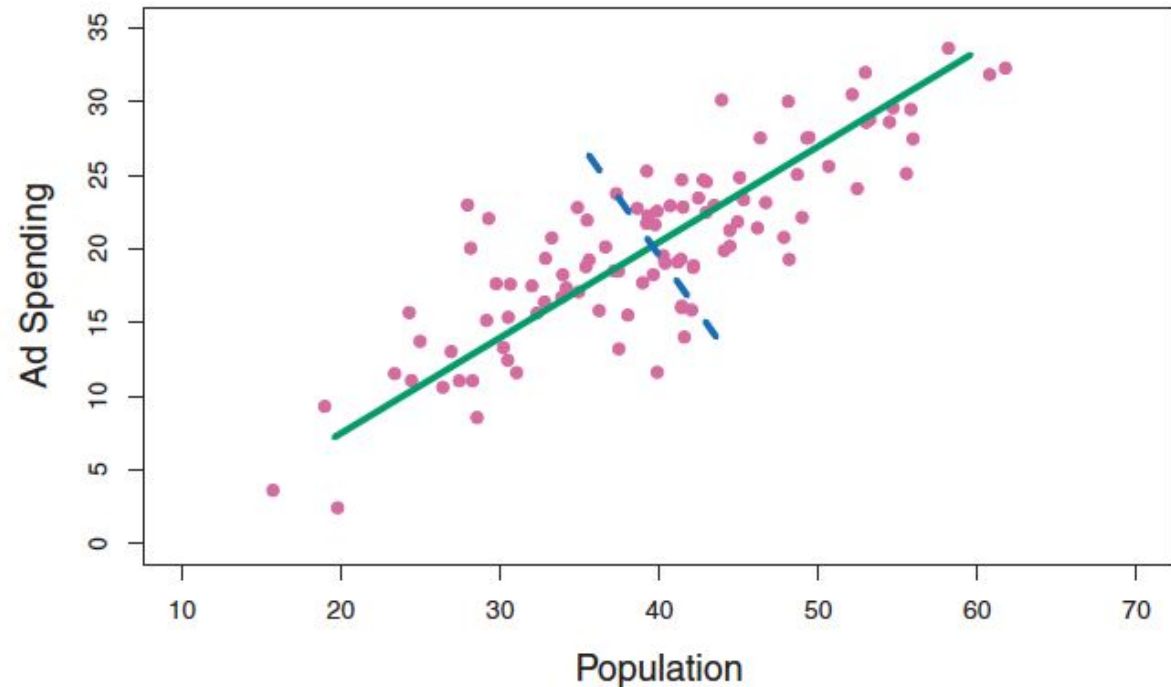
Dimension Reduction Methods

- In this section, we will explore a class of approaches that transform the predictors and then fit a least square model using the transformed variables.
- We refer these techniques as dimension reduction methods.
- The term dimension reduction originates from the fact that using this approach, we can reduce the problem of estimating the $p + 1$ coefficients $\beta_0, \beta_1, \dots, \beta_p$ to a simpler problem of estimating the $M + 1$ coefficients $\theta_0, \theta_1, \dots, \theta_M$ where $M < p$.
- Hence the dimension of the problem has been reduced from $p + 1$ to $M + 1$.



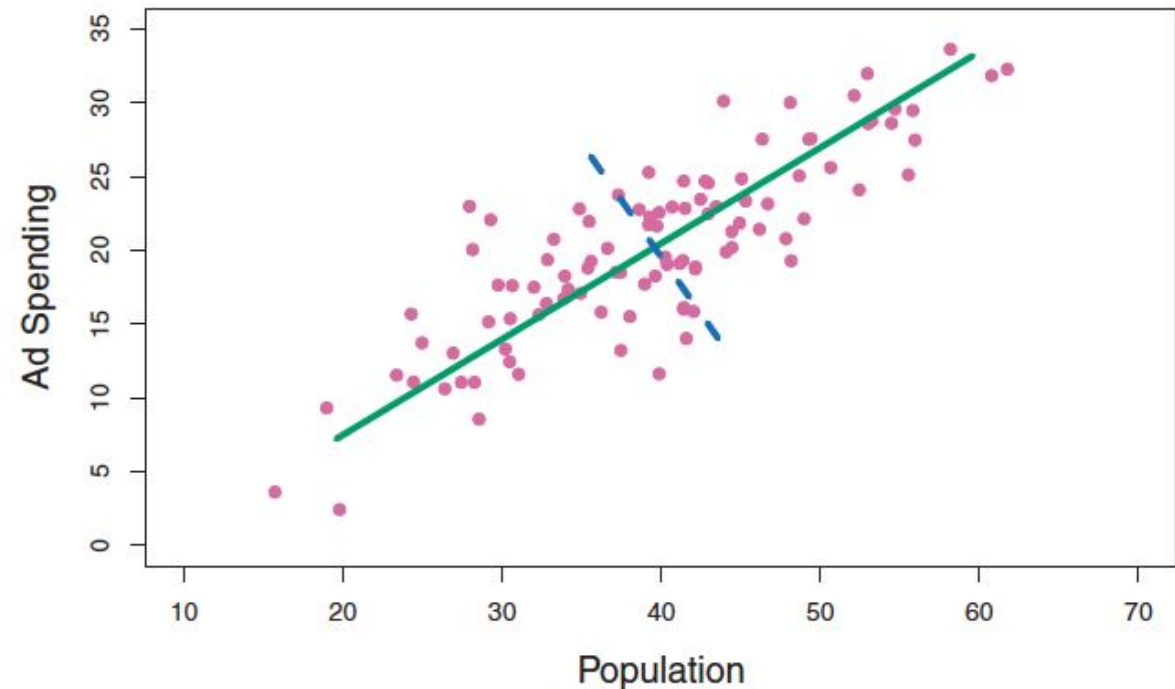
Principal Component Analysis

- We use the PCA to reduce the dimension of a $n \times p$ data matrix $\mathbf{X}$.
- The first principal component direction of the data is that along which the observations vary the most.
- For example, in the figure – 8, we have demonstrated the population size (pop) in tens of thousands of people and ad spending for a particular company (ad) in thousands of dollars for 100 cities.



Principal Component Analysis

- The first principal component direction of the data is represented by a green solid line.
- This is the direction along which the variability in the data is the greatest.
- If we project the 100 observations on this line, then the resulting projected observations would have the largest possible variance.
- Projecting the observations onto any other line would yield projected observations with lower variance



Principal Component Analysis

- In some datasets, the variables correlate with each other.
- For example, the speed and the fuel consumption will be heavily correlated in an automobile.
- While for some application, we can find these correlations, but machine learning algorithms typically do not need redundant information.
- The aim of the PCA is to find the principal components, which are the variables that do not correlate with each other and explain the information, specifically the variance of the dataset.
- It means that we can capture most of the information in a dataset in fewer features.

Thanks

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